Does Medical Pretraining Change What a Chest-Radiograph Classifier Needs? A Frozen-Protocol Comparison on CheXpert

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Purpose

Chest-radiograph classifiers are conventionally reported by AUC or overall accuracy, metrics that can conceal a severe, clinically disqualifying failure mode on a long-tailed, multi-label diagnostic task: a model can score highly on accuracy while missing most true positives. Large-scale medical pretraining — exemplified by Ark+ [1], trained across six chest-radiograph datasets — is increasingly used to build chest X-ray foundation models, but it remains unclear whether such pretraining actually changes what a classifier *needs*: how long it must train before overfitting, how well-calibrated its raw outputs are, and whether it still benefits from long-tail-aware loss design once large-scale pretraining is present. This work asks those three questions directly, under a single frozen protocol, rather than proposing a new architecture or loss.

Methods

We compare six chest-radiograph classification architectures on CheXpert [2]: Ark+ (Swin-Large [5], large-scale medical pretraining), an architecture-matched no-pretraining control (Swin-Large, random init — isolating the effect of pretraining with architecture held fixed), Swin Transformer V2 Base [6] (ImageNet-22k), DenseNet-121 [7] (ImageNet-1k), RAD-DINO [9] (DINOv2 ViT-B/14), and ConvNeXt V2 Base [8] (ImageNet-22k). Ark+'s three original task-specific components (per-class uncertain-label strategy, class-weighted loss, co-occurrence regularization) are stripped so that all six models are evaluated under one identical recipe, listed in Table 1.

Table 1: Frozen training protocol, identical across all six architectures.

Setting	Value
Input resolution	320×320 (native CheXpert-small short side)
Optimizer	AdamW, cosine schedule, 1 warmup epoch, lr 5×10^{-5}, wd 0.05
Epochs	30, checkpoint selected by best *validation* macro-AUC
Validation set	Official CheXpert `valid.csv` (234 studies)
Test set	Patient-disjoint held-out (15% of patients), touched once, after training
Uncertain labels	U-Zeros (headline); U-Ones as an appendix sensitivity check
Seeds	2 for headline results (13, 42), 1 for screening
Metrics	Macro-AUC, F1, sensitivity, accuracy, MCC, AP, ECE

Three questions are asked of this shared design: (a) *does pretraining delay overfitting*, isolated cleanly via the Ark+/Swin-Large architecture-matched pair (pretraining is the only variable); (b) *does pretraining produce a better-calibrated model*, and does per-label operating-point (threshold, selected by Youden's J statistic [10]) policy close the accuracy–sensitivity gap that raw AUC hides; (c) *do large-scale-pretrained encoders still need long-tail-aware loss*, tested architecture-conditionally across all six models. Threshold policies and temperature calibration [4] are fit on the validation set only and applied once to the held-out test set, avoiding the leakage

that results from tuning a threshold on the same set it is then evaluated on.

Results (preliminary)

Pilot architecture comparison

Prior to finalizing the frozen protocol above, an earlier pilot pass trained five configurations under a shared (though non-frozen) setup: 50 epochs, single seed, evaluated on the same internal test split for all five. Swin-Large with random initialization (the same no-pretraining architecture used as the control for question (a), but not yet run under the current frozen settings), DenseNet-121, RAD-DINO, Swin Transformer *V1* Base (superseded by V2 in the frozen protocol above), and ConvNeXt V2 Base. Table 2 reports these; results are *not* directly comparable to Ark+'s frozen-protocol result below, since epoch count, seed count, and protocol version all differ.

Table 2: Preliminary pilot comparison, non-frozen protocol (50 epochs, 1 seed, shared internal test split). Not comparable to Table 3.

	AUC	MCC	AP	F1	Recall	Accuracy
Swin-Large (random init)	0.880	0.565	0.691	0.665	0.788	0.810
DenseNet-121	0.873	0.531	0.687	0.646	0.856	0.778
RAD-DINO	0.877	0.557	0.704	0.670	0.828	0.800
Swin Transformer V1 Base	0.878	0.560	0.700	0.671	0.808	0.804
ConvNeXt V2 Base	0.879	0.548	0.712	0.661	0.858	0.791

Swin-Large led this single-seed pilot on AUC, MCC, and accuracy, but showed the *lowest* sensitivity of the five (0.788) — the same accuracy/sensitivity dissociation examined in depth for Ark+ below. The top three architectures on AUC (0.877–0.880) sit within roughly one seed's expected noise of each other, so this ranking is suggestive rather than conclusive; it will be re-tested under the frozen protocol and 2-seed headline budget above.

Ark+ headline result

Ark+'s headline result is complete under the frozen protocol: Table 3 reports held-out test performance across the five CheXpert competition classes (Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion), mean $\pm$ std over 2 seeds.

Table 3: Ark+ held-out test performance, mean $\pm$ std over 2 seeds (13, 42).

	AUC	F1	Sensitivity	Accuracy	MCC	AP	ECE
Ark+	0.806 ± 0.003	0.472 ± 0.005	0.783 ± 0.004	0.705 ± 0.008	0.369 ± 0.006	0.480 ± 0.004	0.013 ± 0.002

The Sensitivity column already reflects the validation-tuned Youden's-J operating point, not the naive 0.5 threshold. Figure 1 shows why that choice matters: at the naive 0.5 threshold, Ark+ reaches 0.86 accuracy but only 0.29–0.32 mean sensitivity — missing roughly seven in ten true positives despite a high headline accuracy. A validation-tuned threshold (Youden's J or F1-optimal) recovers sensitivity to 0.73–0.84 at a real but bounded accuracy cost, at *unchanged* AUC, since AUC is threshold-independent. Both seeds show this pattern with

near-identical magnitude, indicating it is not an artifact of one run.

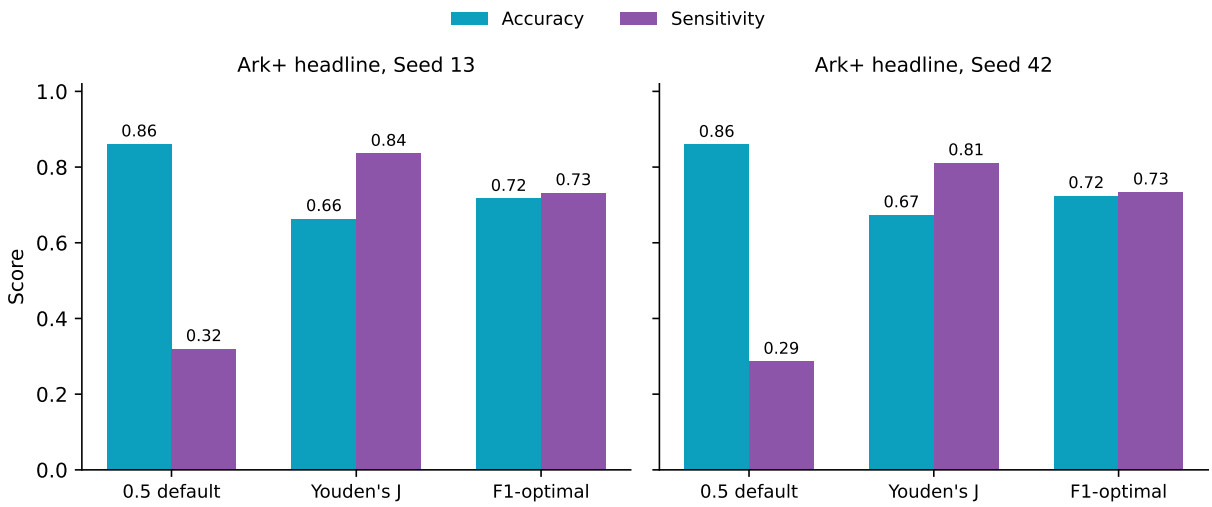


Figure 1: Accuracy vs. sensitivity by threshold policy, Ark+ held-out test, both headline seeds. Thresholds and calibration are fit on validation only and applied once to test.

A single global temperature-scaling calibration, also fit on validation, did *not* improve held-out calibration in this preliminary result (ECE increased from ≈0.011–0.015 to ≈0.070 after scaling); the fitted temperature ($T \approx 1.6$) over-corrected a model whose raw outputs were already reasonably well-calibrated on this split. We report this as a genuine negative finding, likely attributable to the small (234-image) official validation set, rather than omit it.

A specific, targetable weakness. Per-class results show Atelectasis and Consolidation lagging on AP (0.29, 0.17 vs. 0.51–0.81) and F1 (0.35, 0.20 vs. 0.46–0.75), despite comparable AUC (0.71, 0.73 vs. 0.86–0.88) — a precision/recall problem, not a ranking problem, consistent with class imbalance.

First intervention, already tested. Class-weighted BCE — one of Ark+'s original components stripped for this comparison — was reintroduced as a quick screening-tier test (1 seed, 20% subset), with `pos_weight` set from the real per-class balance. It already improves the model on the classes it targets: at the naive 0.5 threshold, F1 rose substantially for both (Atelectasis $0.02 \rightarrow 0.33$, Consolidation $0.01 \rightarrow 0.19$). AP moved slightly down for every class (-0.01 to -0.02), so this reflects a shifted operating point rather than confirmed better ranking, and may overlap with what threshold-tuning alone (Figure 1) already recovers without retraining. Screening-tier, 1-seed signal, not yet confirmed at headline tier.

Remaining planned interventions, by cost: (i) the asymmetric/focal loss analysis in question (c); (ii) a larger, patient-disjoint threshold-tuning split; (iii) DenseNet-121/Ark+ logit fusion, if DenseNet-121 shows a complementary error pattern; and (iv) class-balanced oversampling.

Training for the remaining five architectures under the current frozen protocol — including the Ark+/Swin-Large controlled pretraining comparison central to question (a) — and the architecture-conditional loss-design analysis for question (c) are in progress and will be presented at the conference.

Novel and Breakthrough Aspects

To our knowledge this is among the first independent evaluations of Ark+ (published June 2025) that isolates the effect of its large-scale pretraining from its architecture via a matched no-pretraining control, rather than

comparing it only against unrelated architectures. The threshold-policy result in Figure 1 makes a specific, quantitative version of a widely-suspected but rarely-isolated claim: that headline accuracy on CheXpert-scale long-tailed data can be nearly independent of clinically-relevant sensitivity, and that this gap is substantially — though not fully — closeable without any additional training, purely through validation-tuned operating-point selection.

Related Work

CheX-DS [3] ensembles DenseNet and a Swin Transformer with weighted BCE plus asymmetric loss on NIH ChestX-ray14 (83.76% average AUC). We differ in dataset (CheXpert vs. ChestX-ray14), scope (six architectures under one frozen protocol vs. one ensemble), and framing: we claim no novelty for ensembling or asymmetric loss; question (c) instead asks, architecture-conditionally, whether large-scale pretraining already absorbs the long-tail imbalance such losses target. The planned DenseNet-121/Ark+ fusion (above) stays a subordinate diagnostic, not a headline contribution, given CheX-DS's prior ensemble result.

Conclusion

Under identical protocol conditions, Ark+ shows a large, real gap between headline accuracy and clinically-relevant sensitivity that validation-tuned thresholding substantially recovers at no AUC cost — but naive single-temperature calibration on a small validation set did not help, and may hurt. Whether large-scale medical pretraining changes training length, calibration behavior, or long-tail loss requirements relative to architecture-matched and ImageNet-pretrained baselines is being tested directly via the controlled comparisons above; full six-architecture results will be presented.

Disclosure of Prior or Concurrent Submissions

[FILL IN — required field. State here whether this work, or any subset of it, has been submitted, presented, or published elsewhere, or state that it has not.]

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